Lecture 5 recap

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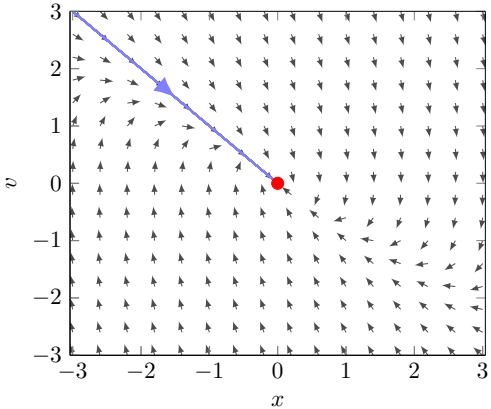
$$\begin{pmatrix} x \\ v \end{pmatrix} = C_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 e^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \quad \boldsymbol{y}' = \mathbf{A} e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$= -e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



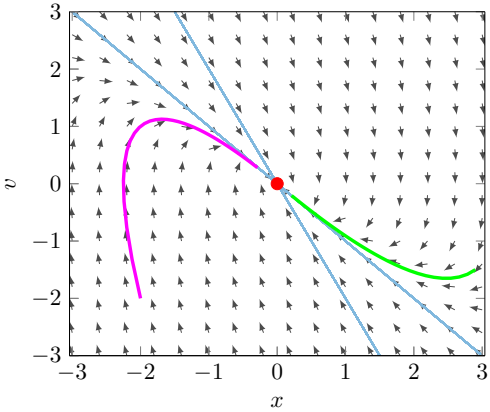
Lecture 6

In Lecture 6, we will:

1. Learn how to sketch trajectories when the initial condition does not lie on an eigenvector
2. Connect the long-term behaviours of solutions to the stabilities of critical points
3. Learn how to solve inhomogeneous systems by diagonalising the matrix $\mathbf{A}$

Sketching phase portraits - curved orbits

What happens if the initial condition is NOT along one of the eigenvectors?



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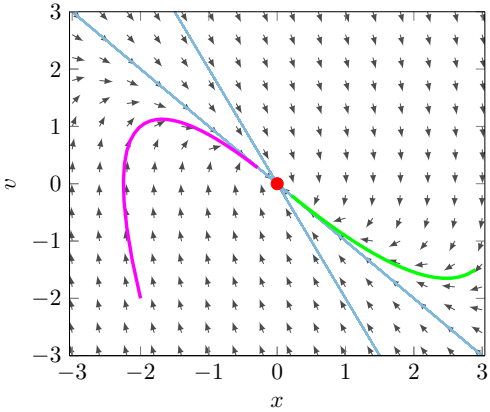
- Linear transformation stretches/compresses/flips AND rotates initial condition.



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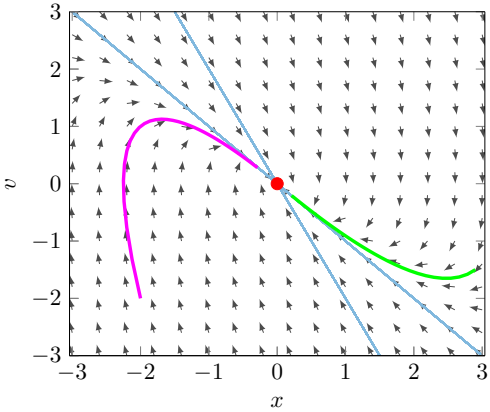

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What happens if the initial condition is NOT along one of the eigenvectors?

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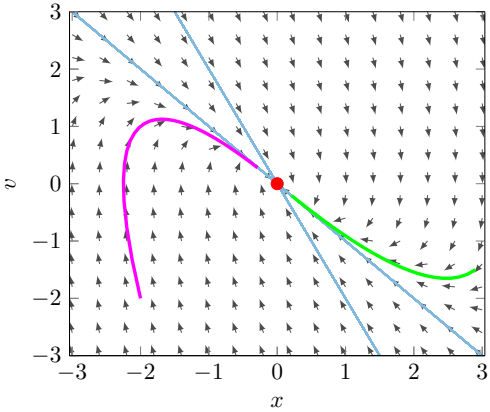
- Solution is a parametric curve, with arrow pointing in direction of increasing t.
- Solution for large times 'grounded' by eigenvectors.



Sketching phase portraits - curved orbits

Back to Problem 1.2: Recall that

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$$\Rightarrow \quad \boldsymbol{y}_0 = C_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \text{for} \quad C_1, C_2 \neq 0$$

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- This tell us that curved trajectories approach critical point along the first eigenvector as $t \to \infty$.

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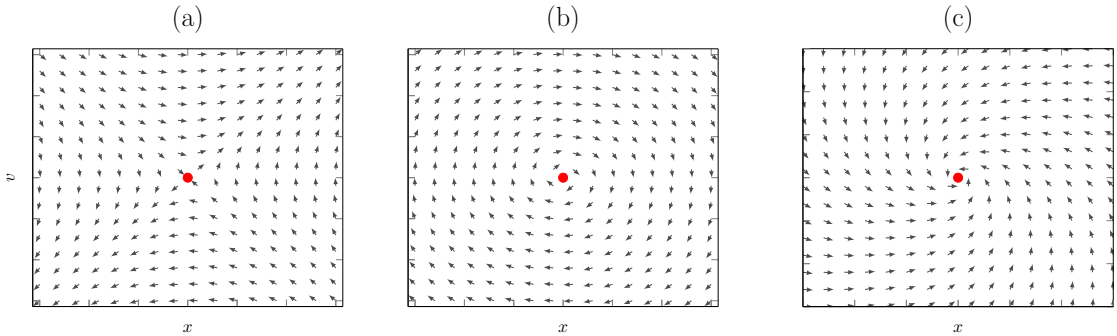
  - This means that it is possible for a solution to start arbitrarily close to the critical point and eventually leave the neighbourhood of that point.
- The origin $\boldsymbol{y} = \mathbf{0}$ is stable if the real parts of the eigenvalues are negative or zero. In particular, it is asymptotically stable if the real parts of the eigenvalues are negative.

Exercise 1.13: Which of the following critical points is stable?



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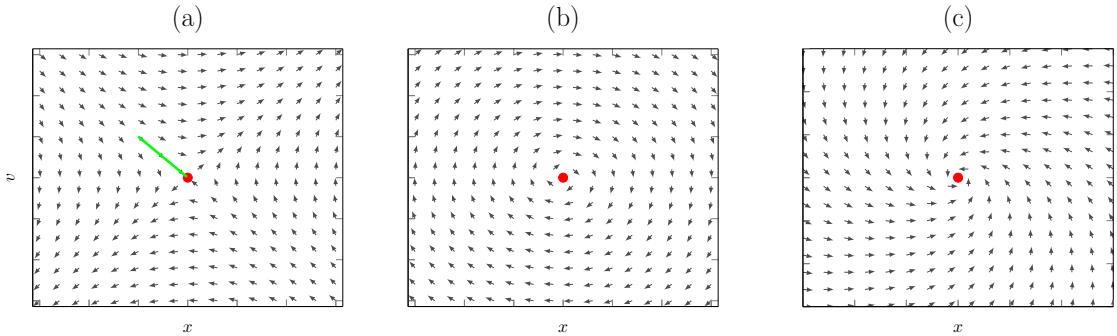

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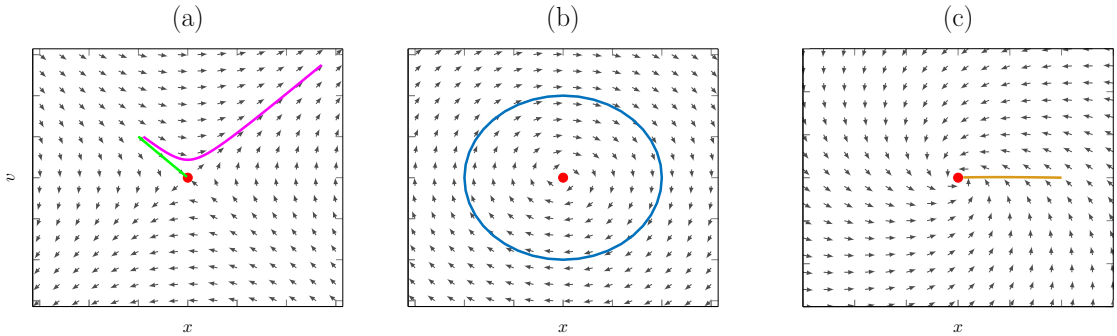

Case study 2: Wildlife conservation

- As far back as 15 million years ago, the greater bilby (*Macrotis lagotis*) roamed and inhabited 70 per cent of the Australian mainland. Today, due to predation by animals like dingoes, along with competition for food from rabbits and loss of habitat from urbanisation, this iconic marsupial is listed as a critically endangered species (Blumstein et al (2019) Phil. Trans. R. Soc.B.)
- Understanding the coevolutionary dynamics of predator-prey interactions through mathematical models, specifically **IVPs - inhomogeneous systems**, is important in predicting future populations and developing effective management strategies that allow the population of threatened species like the bilby to recover.

Diagonalisable systems

Revision

If $\mathbf{A}$ be a diagonalisable matrix, then $\mathbf{A} = \mathbf{S}\,\mathbf{D}\,\mathbf{S}^{-1}$ where

- $\mathbf{D}$ is a diagonal matrix whose entries are the eigenvalues
- and the columns of $\mathbf{S}$ contain the eigenvectors.

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- We have decoupled the system.
- It can be solved as n independent first-order linear ODEs.

Exercise 1.14

Use the method of diagonalisation to solve the following system of ODEs:

$$x' - y = -\mathrm{e}^{-t}$$
$$y' + 2x + 3y = \mathrm{e}^{-t}$$

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First we write as a vector equation

$$y = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\boldsymbol{y}' = \underbrace{\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}}_{A} \boldsymbol{y} + \underbrace{\begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix}}_{b}.$$

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The eigenvalues and eigenvectors of the matrix are

$$\lambda_1 = -1 \quad \text{with} \quad \boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \lambda_2 = -2 \quad \text{with} \quad \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Thus

$$\boldsymbol{S} = \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix}$$

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The diagonalised system is

$$\boldsymbol{z}' = \underbrace{\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}}_{D} \boldsymbol{z} + \boldsymbol{S}^{-1} \begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix}$$

$\underline{z} = S^{-1} \underline{y}$

Thus

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Solve this decoupled system using integrating factors (exercise) to get

$$\boldsymbol{z} = \begin{pmatrix} (C_1 + t)\,\mathrm{e}^{-t} \\ C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

$z_1' = -z_1 + e^{-t}$

$I = e^{\int 1\,dt} = e^{t}$

$(e^{t} z_1)' = 1 \Rightarrow e^{t} z_1 = t + C_1$

$z_1 = t e^{-t} + C_1 e^{-t}$

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$$\Rightarrow \quad \boldsymbol{y} = \boldsymbol{S}\boldsymbol{z} = \begin{pmatrix} -(C_1 + t)\,\mathrm{e}^{-t} - C_2 \mathrm{e}^{-2t} \\ (C_1 + t)\,\mathrm{e}^{-t} + 2C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

Summary of systems ODEs

1. Higher order ODEs can be written as a single *vector-valued* ODE.
2. Constant coefficient homogeneous: use exponential trial functions.
3. Constant coefficient inhomogeneous: use matrix diagonalisation to decouple the system then use an integrating factor on individual ODEs.

When does it fall apart?

- Matrix is not diagonalisable or coefficients are functions of t: learn matrix exponentials!